

E-commerce Sales Analysis Project

This notebook performs end-to-end exploratory data analysis (EDA), customer segmentation, and visualization on an e-commerce dataset.

```
In [17]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [18]: df = pd.read_csv('../data/data.csv', encoding='ISO-8859-1')
df.head()
```

```
Out[18]:
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	01-12-2010 08:26	2.55	17850.0	United Kingdom
1	536365	71053	WHITE METAL LANTERN	6	01-12-2010 08:26	3.39	17850.0	United Kingdom
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	01-12-2010 08:26	2.75	17850.0	United Kingdom
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	01-12-2010 08:26	3.39	17850.0	United Kingdom
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	01-12-2010 08:26	3.39	17850.0	United Kingdom

Basic Info:-

We inspect data types, summary statistics, and basic structure of the dataset to understand its quality.

```
In [19]: df.info()
df.describe()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 541909 entries, 0 to 541908
Data columns (total 8 columns):
#   Column          Non-Null Count  Dtype  
---  -
0   InvoiceNo        541909 non-null  str    
1   StockCode       541909 non-null  str    
2   Description     540455 non-null  str    
3   Quantity        541909 non-null  int64  
4   InvoiceDate      541909 non-null  str    
5   UnitPrice       541909 non-null  float64
6   CustomerID      406829 non-null  float64
7   Country         541909 non-null  str    
dtypes: float64(2), int64(1), str(5)
memory usage: 33.1 MB
```

Out[19]:

	Quantity	UnitPrice	CustomerID
count	541909.000000	541909.000000	406829.000000
mean	9.552250	4.611114	15287.690570
std	218.081158	96.759853	1713.600303
min	-80995.000000	-11062.060000	12346.000000
25%	1.000000	1.250000	13953.000000
50%	3.000000	2.080000	15152.000000
75%	10.000000	4.130000	16791.000000
max	80995.000000	38970.000000	18287.000000

Missing Values Check:-

We check for missing values in the dataset to identify incomplete or null data entries.

This helps us decide whether to drop or impute missing values before analysis.

In [20]: df.isnull().sum()

Out[20]: InvoiceNo 0
StockCode 0
Description 1454
Quantity 0
InvoiceDate 0
UnitPrice 0
CustomerID 135080
Country 0
dtype: int64

Data Cleaning:-

We clean the dataset by:

- Removing missing CustomerID values
- Converting InvoiceDate to datetime format
- Removing invalid transactions (Quantity ≤ 0, UnitPrice ≤ 0)
- Creating a new feature: TotalPrice

In [21]: df = df.dropna(subset=['CustomerID'])
df['InvoiceDate'] = pd.to_datetime(df['InvoiceDate'], dayfirst=True)
df = df[(df['Quantity'] > 0) & (df['UnitPrice'] > 0)]
df['TotalPrice'] = df['Quantity'] * df['UnitPrice']

```
df.head()
```

Out[21]:	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	TotalPrice
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	17850.0	United Kingdom	15.30
1	536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom	20.34
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	17850.0	United Kingdom	22.00
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom	20.34
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom	20.34

Total Revenue:-

We calculate the total revenue generated by the e-commerce store.

```
In [22]: df['TotalPrice'].sum()
```

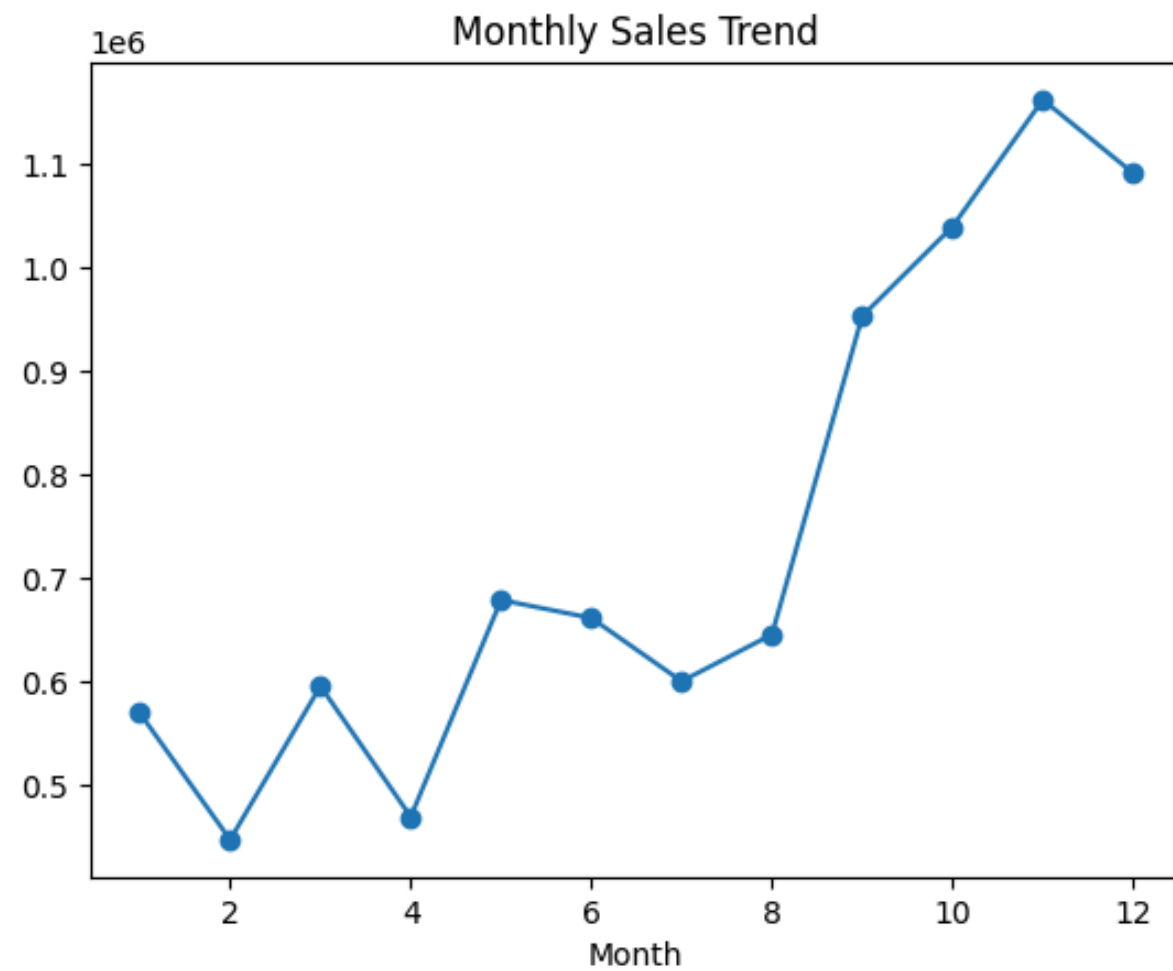
```
Out[22]: np.float64(8911407.904)
```

Monthly Sales Trend:-

We analyze how sales vary across different months to identify seasonal patterns.

```
In [23]: df['Month'] = df['InvoiceDate'].dt.month
monthly = df.groupby('Month')['TotalPrice'].sum()

monthly.plot(marker='o')
plt.title('Monthly Sales Trend')
plt.show()
```

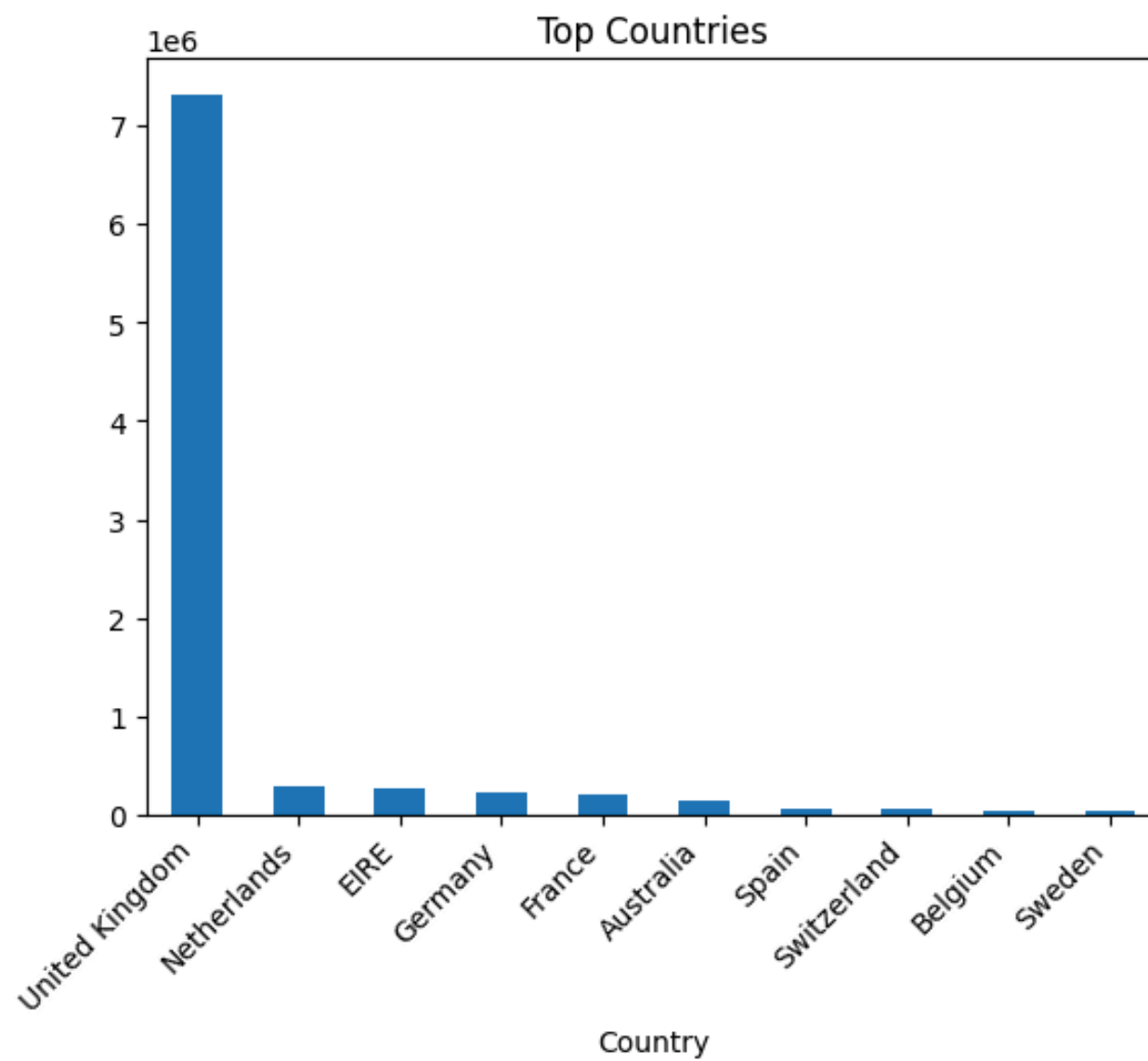


Top Countries by Revenue:-

We identify which countries contribute the highest revenue to the business.

```
In [24]: top_countries = df.groupby('Country')['TotalPrice'].sum().sort_values(ascending=False).head(10)

top_countries.plot(kind='bar')
plt.title('Top Countries')
plt.xticks(rotation=45, ha='right')
plt.show()
```

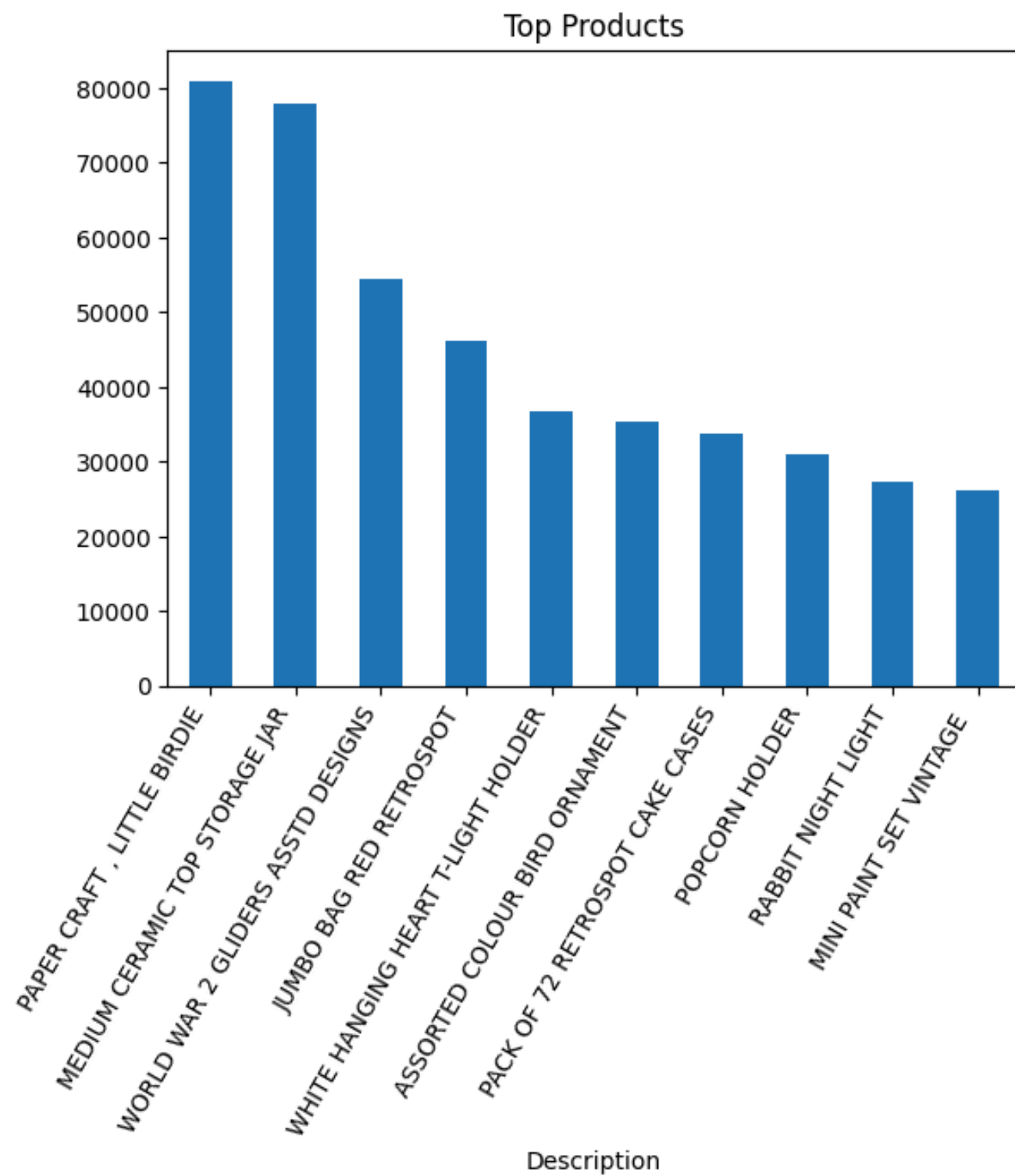


Top Selling Products:-

We find the most frequently sold products to understand customer demand.

```
In [25]: top_products = df.groupby('Description')['Quantity'].sum().sort_values(ascending=False).head(10)

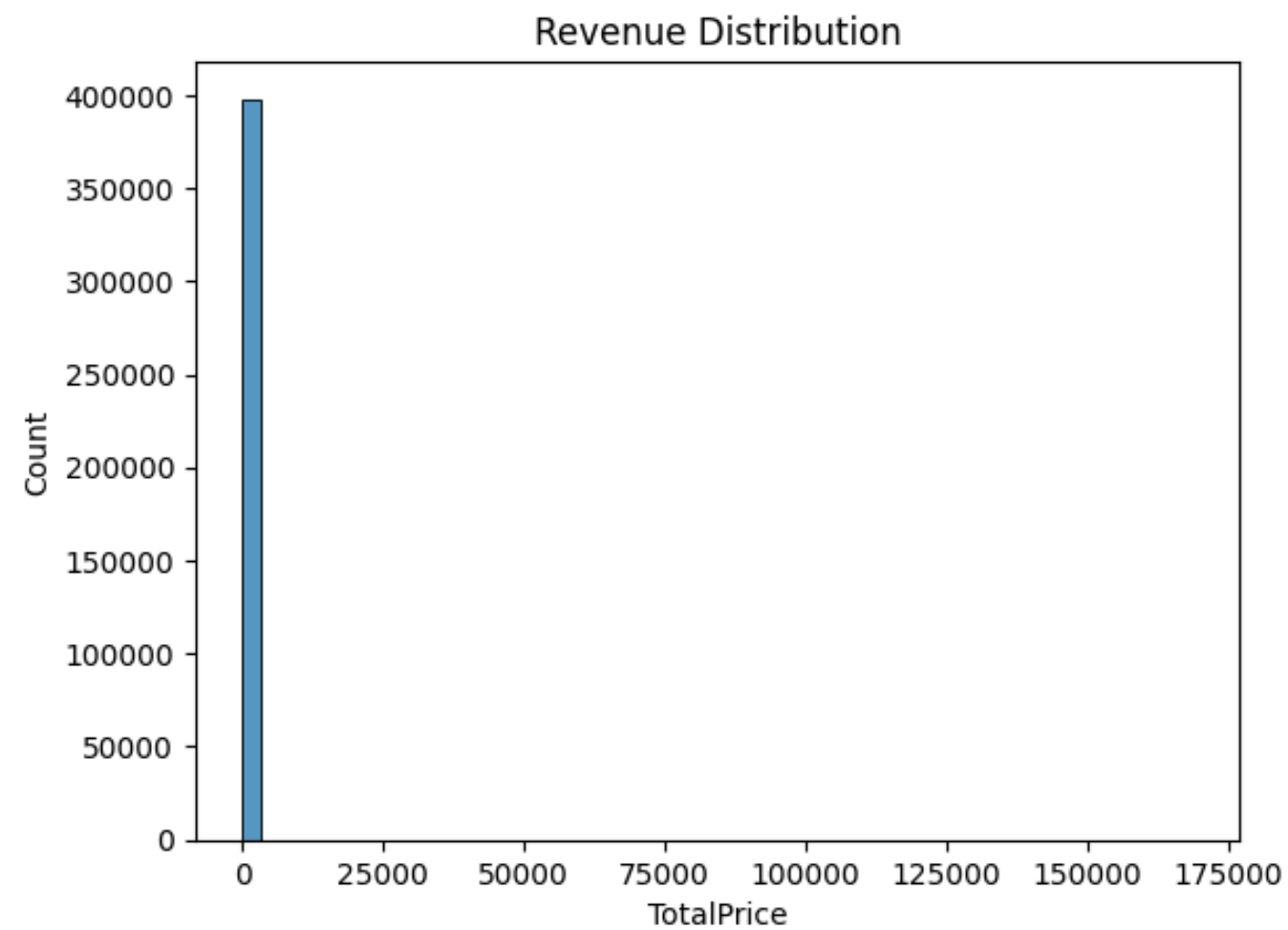
top_products.plot(kind='bar')
plt.title('Top Products')
plt.xticks(rotation=60, ha='right')
plt.show()
```



Revenue Distribution:-

We analyze how transaction values are distributed across all purchases.

```
In [26]: sns.histplot(df['TotalPrice'], bins=50)
plt.title('Revenue Distribution')
plt.show()
```

Customer Segmentation (RFM Analysis):-

We segment customers based on:

- Recency (last purchase)
- Frequency (number of purchases)
- Monetary value (total spending)

This helps identify high-value customers.

```
In [27]: snapshot = df['InvoiceDate'].max() + pd.Timedelta(days=1)

rfm = df.groupby('CustomerID').agg({
    'InvoiceDate': lambda x: (snapshot - x.max()).days,
    'InvoiceNo': 'count',
    'TotalPrice': 'sum'
})

rfm.columns = ['Recency', 'Frequency', 'Monetary']
rfm.head()
```

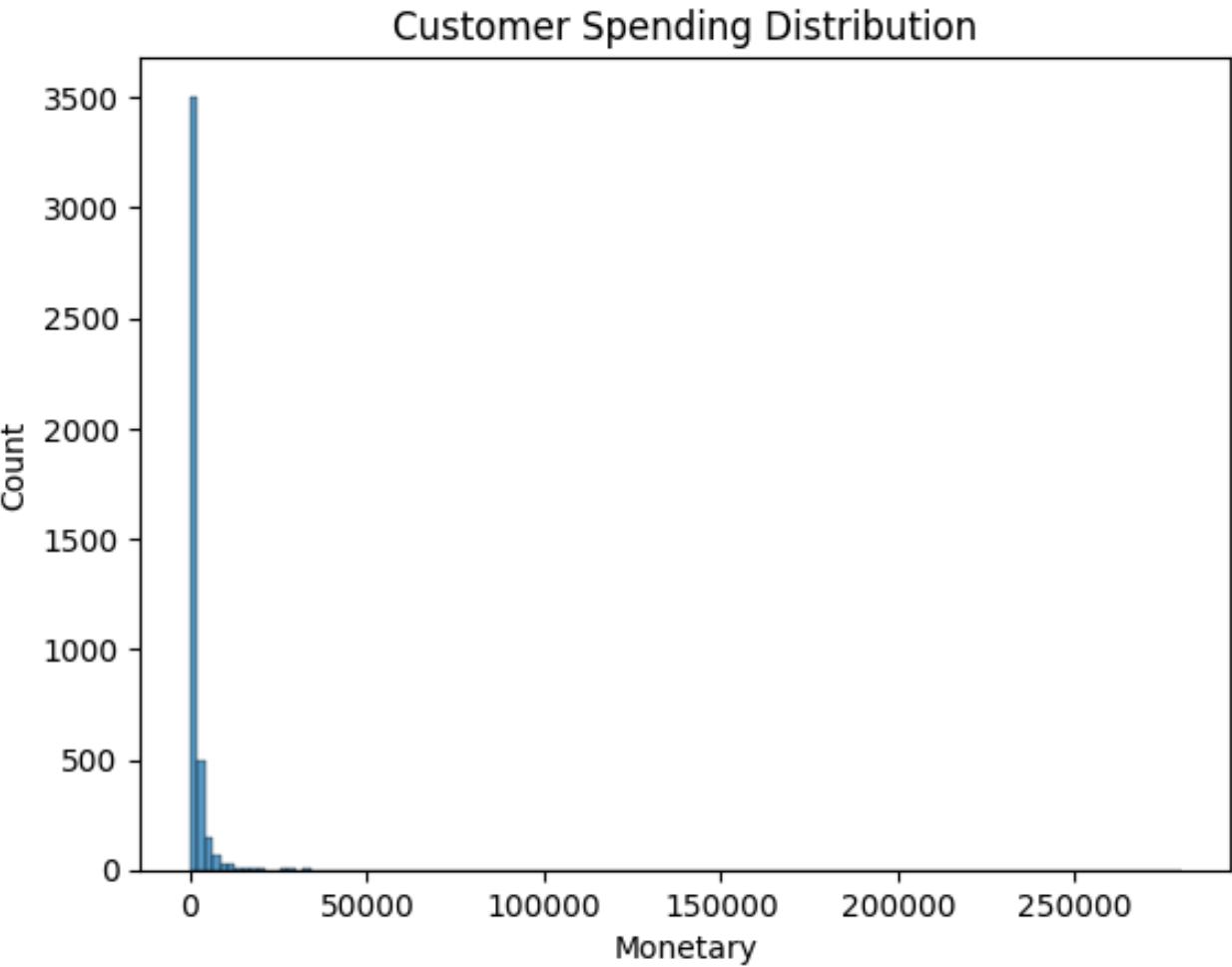
Out[27]:

	Recency	Frequency	Monetary
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CustomerID			
12346.0	326	1	77183.60
12347.0	2	182	4310.00
12348.0	75	31	1797.24
12349.0	19	73	1757.55
12350.0	310	17	334.40

RFM Visualization:-

```
In [28]: sns.histplot(rfm['Monetary'])
plt.title('Customer Spending Distribution')
plt.show()
```



🧠 Key Business Insights

- 💰 Only a small segment of customers contributes the majority of revenue (Pareto Principle observed)
- 🌐 United Kingdom is the dominant market in terms of sales

- 📦 A limited number of products generate most of the revenue
- 📅 Strong seasonal trends observed in sales performance
- 👤 RFM analysis effectively identifies loyal and high-value customers

Conclusion

This analysis provides actionable business insights that can support strategic decision-making in:

- 📈 Marketing optimization
- 🎯 Customer targeting and retention
- 📦 Inventory and supply chain planning
- 💰 Revenue growth and forecasting